

Hypothesis Testing

Q1. What is a null hypothesis (H_0) and why is it important in hypothesis testing?

Ans. The Null Hypothesis

The null hypothesis is a default assumption that there is no effect, no difference or no relationship B/w variables being studied. It represents the status the position that says nothing Interesting is happening here,

Examples:

- A new drug has no effect on blood pressure (vs. the alternative that it does)
- There is no difference in test scores between two teaching methods
- There is no correlation between hours studied and exam performance

Why it is important.

1. It gives you something to test against hypothesis testing works by assuming H_0 is true and then asking how likely is that I'd observe data this extreme by Pure chance? You can't prove something exists directly but you can show the "nothing happening" explanation.
2. It controls false positives (Type I errors) by setting a significance threshold you limit The probability of incorrectly rejecting H_0 when it's actually true. Without this anchor, Researcher could cherry pick results.
3. It makes science falsifiable Following Popper's principle of falsifiability, science advances by *disproving* claims rather than confirming them. H_0 provides a precise, testable claim that evidence can challenge.
4. It enables the p-value framework The p-value is defined as: $P(\text{observing data this extreme} \mid H_0 \text{ is true})$. The null hypothesis is literally baked into the definition — without it, p-values have no meaning.

Q2. What does the significance level (α) represent in hypothesis testing?

Ans. Our courtroom analogy from before? You're the judge and you need to decide

How much evidence is enough to declare someone guilty the significance level

- (a) Is exactly that- you setting your own bar for how much proof you need before
You say "okay something real is going on here"

In Dead Simple Terms

α = the chance you're willing to be wrong.

If $\alpha = 0.05$

The Key Idea

Think of α as a **sensitivity dial**.

- **Lower α (like 0.01)** → You're very hard to convince. You need really strong evidence. Fewer false alarms but you might miss real effects.
- **Higher α (like 0.10)** → You're easier to convince. More sensitive but you'll make more mistakes.

Q3. Differentiate between Type I and Type II errors.

Ans. Type I vs Type II Errors

Let's stick with our courtroom analogy since you already know it well.

The Simple Version

Type I Error = You declared an innocent person guilty

Type II Error = You let an actual criminal go free

That's it. Two different ways justice can fail — and hypothesis testing can fail the same way.

Now in Hypothesis Testing Terms

Type I Error – You reject H_0 but H_0 Was actually true

Type II Error – You kept H_0 but H_0 was actually false

Type I Error → You said "*the drug works!*" but it actually **doesn't**. You got excited over nothing. A false alarm.

Type II Error → You said "*the drug doesn't work*" but it actually **does**. You just threw away a life-saving medicine.

Type I = You cried wolf when there was no wolf. Type II = There WAS a wolf and you said nothing.

Q4. Explain the difference between a one-tailed and two-tailed test. Give an example of each.

Ans. One-Tailed vs Two-Tailed Tests

Let's build on what you already know. You're testing whether something has an effect — but now the question is which direction are you looking for that effect?

The Simple Analogy

you're watching a door and waiting for someone suspicious to walk through.

- **Two-tailed** → You're watching **both sides** — someone could come from the left OR the right. You don't know which direction.
- **One-tailed** → You **only watch one side** — you already know if anyone comes, they're coming from the left.

Example

a new study method. She wants to know if it changed test scores — **better OR worse**.

- H_0 = The method made no difference
- H_1 = The method made **some** difference (could be higher or lower)

Q5. A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim.

Ans. **Step 1** — Set Up Your Hypotheses

H_0 (Null): The average time really is 10 minutes → $\mu = 10$

H_1 (Alternative): The average time is NOT 10 minutes → $\mu \neq 10$

We use $\neq$ here because we're checking if the company's claim is wrong in **either direction** — so this is a **two-tailed test**.

Step 2 — Choose Your Test

Since:

- Sample size is **small** ($n = 9$, less than 30)
- We **don't know** the population standard deviation
- We only have the sample standard deviation

We use a **t-test** — not a z-test.

Step 3 — Calculate the t-Score

The formula is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Plugging in our numbers:

$$\begin{aligned} t &= \frac{12 - 10}{3/\sqrt{9}} \\ t &= \frac{2}{3/3} \\ t &= \frac{2}{1} \\ t &= 2.0 \end{aligned}$$

Our t-score is 2.0 — this tells us our sample result is 2 standard errors away from the company's claim.

Step 4 — Find the Critical Value

- Degrees of freedom = $n - 1 = 9 - 1 = 8$
- Two-tailed test at $\alpha = 0.05$
- From the t-table → critical value = **±2.306**

Final Answer is

t- calculated – 2.0

t- critical - ±2.306

Decision - Fail to reject H_0

Conclusion – Not enough evidence to challenge the company's claim

Q6. When should you use a Z-test instead of a t-test?

Ans. You already used a t-test in Q5 because the sample was small. But when do you switch to a **Z-test**? Let's break it down in plain human language.

The Simple Analogy

Think of it like asking for directions in an unfamiliar city.

- **Z-test** = You have **Google Maps** — you know exactly where everything is, full information, very confident.
- **T-test** = You're asking a **local stranger** — they give you a good estimate, but there's some uncertainty baked in.

Use Z-test

A factory tests 500 light bulbs and wants to check if average lifespan matches the claimed 1000 hours. They also know from years of records that the population standard deviation is 50 hours.

- Large sample ✓
- Known population standard deviation ✓
- Z-test is the right call

Use T-test

A doctor tests a new pill on 8 patients and measures blood pressure. He only has the sample standard deviation from those 8 people.

- Small sample ✓
- Unknown population standard deviation ✓
- T-test is the right call

Z-test = large sample + you know the population standard deviation

T-test = small sample + you only have the sample standard deviation

When in doubt — use the t-test. It's the safer, more honest choice.

Q7. The productivity of 6 employees was measured before and after a training program.

Employee	Before	After
1	50	55
2	60	65
3	58	59
4	55	58
5	62	63
6	56	59

At $\alpha = 0.05$, test if the training improved productivity.

Ans. **Step 1 — Set Up Hypotheses**

H₀: Training made no difference $\rightarrow \mu_d = 0$

H₁: Training improved productivity $\rightarrow \mu_d > 0$

We use $>$ because we're specifically checking if scores went **UP** after training — so this is a **one-tailed test**.

Step 2 — Calculate the Mean Difference ($\bar{d}$)

$$\bar{d} = \frac{5 + 5 + 1 + 3 + 1 + 3}{6} = \frac{18}{6} = 3$$

Average improvement = 3 points per employee

Step 3 — Calculate Standard Deviation of Differences

First find each difference minus the mean, then square it:

Sum of squares = $4 + 4 + 4 + 0 + 4 + 0 = 16$

$$s_d = \sqrt{\frac{16}{6 - 1}} = \sqrt{\frac{16}{5}} = \sqrt{3.2} = 1.789$$

Step 4 — Calculate the t-Score

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{3}{1.789/\sqrt{6}} = \frac{3}{1.789/2.449} = \frac{3}{0.730} = 4.11$$

Our t-score = 4.11

Step 5 — Find the Critical Value

- Degrees of freedom = n - 1 = 6 - 1 = 5
- One-tailed test at α = 0.05
- From t-table → critical value = 2.015

Final summary.

Mean difference (d) – 3.0

Standard deviation – 1.789

t-calculated – 4.11

t-critical – 2.015

decision – Reject H₀

Conclusion – Training significantly improved productivity.

Q8. A company wants to test if product preference is independent of gender

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

At α = 0.05, test independence

Ans. **Step 1 — Set Up Hypotheses**

H₀: Product preference is **independent** of gender (no relationship)

H₁: Product preference **depends** on gender (there is a relationship)

Step 2 — Calculate Expected Values

Expected value formula:

$$E = \frac{RowTotal \times ColumnTotal}{GrandTotal}$$

Step 3 — Compare Observed vs Expected

Already you can see — the numbers are very different. Males prefer A more than expected, Females prefer B more than expected. But let's let the math confirm it.

Step 4 — Calculate Chi-Square Value

Formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2=5.0+3.33+5.0+3.33=16.66$$

Step 5 — Find Critical Value

- Degrees of freedom = (rows - 1) × (columns - 1) = (2-1) × (2-1) = **1**
- $\alpha = 0.05$
- From Chi-Square table → critical value = **3.841**

Final summary

χ^2 calculated – 16.66

χ^2 critical – 3.841

degrees of freedom – 1

Decision – Reject H_0

Conclusion - Product preference depends on gender

DRIVE LINK : https://drive.google.com/drive/folders/1ZlFjacMcn7A-Gi0Qmzbq18jmXQvYluqH?usp=drive_link